

Probability & Statistics (MT-2005)

Descriptive Statistics

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Measures of Dispersion

Motivation

Consider the datasets of two corporations who have hired 10 graduates. The starting salaries for each graduate are shown.

Starting Salaries for Corporation A (in thousands of dollars)

Salary	41	38	39	45	47	41	44	41	37	42
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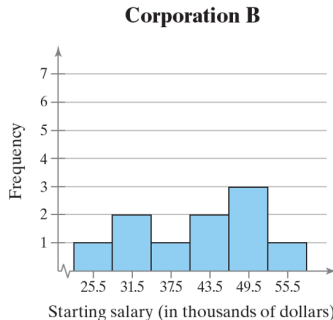
Starting Salaries for Corporation B (in thousands of dollars)

Salary	40	23	41	50	49	32	41	29	52	58
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Both data sets have a mean of 41.5, a median of 41, and a mode of 41. And yet the two sets differ significantly. The salaries of the graduates in Corporation B vary much more than those on Corporation A.

Motivation (cont'd.)

As can be seen in the figures, the starting salaries for Corporation B are more spread out/dispersed than those for Corporation A.



To describe that difference quantitatively, we use a descriptive measure that indicates the amount of dispersion or variation in a data set. Such descriptive measures are referred to as measures of dispersion or measures of variation.

Definition: Measures of dispersion describe the spread or variability in a dataset.

Why is it important?

- Helps understand the degree of variation in the data.
- Complements measures of central tendency by providing additional insights.
- Crucial for identifying consistency and detecting outliers in the data.

Absolute and Relative Measures of Dispersion

Absolute vs. Relative Measures of Dispersion

Absolute Measures of Dispersion:

- Measure the variability of data in the original units.
- Examples: Range, Mean Deviation, Variance, Standard Deviation, Interquartile Range (IQR).
- Provide the actual magnitude of dispersion.

Relative Measures of Dispersion:

- Measure variability relative to a central value (e.g., mean) or as a ratio/percentage.
- Examples: Coefficient of Variation (CV), Relative Range, Coefficient of Range, Quartile Coefficient of Dispersion.
- Useful for comparing dispersion across datasets with different units or scales.

Absolute Measures of Dispersion

Range

Definition: The difference between the maximum and minimum values in the dataset.

Formula:

$$\text{Range} = \text{Maximum Value} - \text{Minimum Value}$$

Characteristics:

- Simple and easy to compute.
- Highly sensitive to outliers.
- Does not provide information about data distribution between the extremes.

Example: Dataset: $\{5, 10, 15, 20, 25\}$

$$\text{Range} = 25 - 5 = 20$$

Interquartile Range (IQR)

Definition: The difference between the third quartile (Q_3) and the first quartile (Q_1).

Formula:

$$\text{IQR} = Q_3 - Q_1$$

where:

- Q_1 : First Quartile (25% of the data below this value)
- Q_3 : Third Quartile (75% of the data below this value)

Characteristics:

- Represents the range of the middle 50% of the data.
- Robust to outliers and extreme values.
- Useful for identifying the spread in skewed distributions.

Example: Dataset: $\{2, 4, 6, 8, 10, 12, 14\}$ $Q_1 = 4$, $Q_3 = 12$

$$\text{IQR} = 12 - 4 = 8$$

Mean Deviation (MD)

The **Mean Deviation** is a measure of dispersion that indicates the average absolute deviation of data points from the central tendency (either the mean or the median). It provides a measure of how spread out the values in a data set are around the center.

Formula for Mean Deviation:

- **When using the mean:**

$$\text{Mean Deviation} = \frac{1}{n} \sum_{i=1}^n |x_i - \bar{x}|$$

where:

- x_i = each data point,
- $\bar{x}$ = mean of the data,
- n = number of data points,
- $|x_i - \bar{x}|$ = absolute deviation of each data point from the mean.

- **When using the median:**

$$\text{Mean Deviation} = \frac{1}{n} \sum_{i=1}^n |x_i - \text{median}|$$

where the median represents the middle value of the data set.

Characteristics of Mean Deviation:

- It is less sensitive to extreme values compared to variance or standard deviation.
- It gives a direct understanding of the spread of the data.
- It can be used with both the mean and the median, depending on the nature of the data.

Definition: The average of the squared deviations from the mean.

Formula:

$$\text{Variance} = \sigma^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$$

Characteristics:

- Expressed in squared units of the data.
- Useful for measuring total variability.
- Heavily influenced by outliers.

Example: Dataset: $\{2, 4, 6, 8\}$, Mean: $\bar{x} = 5$

$$\text{Variance} = \frac{(2 - 5)^2 + (4 - 5)^2 + (6 - 5)^2 + (8 - 5)^2}{4} = \frac{9 + 1 + 1 + 9}{4} = 5$$

Definition: Covariance measures how two variables change together. It indicates the direction of the linear relationship between variables.

Formula:

$$\text{Covariance (Cov)} = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

Key Points:

- Positive covariance: Variables increase together.
- Negative covariance: One variable increases as the other decreases.
- Zero covariance: Variables are uncorrelated.

Definition: Correlation measures the strength and direction of the linear relationship between two variables. It is the normalized version of covariance.

Formula:

$$\text{Correlation } (r) = \frac{\text{Cov}(X, Y)}{\sigma_X \cdot \sigma_Y}$$

Key Points:

- Correlation values range from -1 to 1 .
- $r = 1$: Perfect positive linear relationship.
- $r = -1$: Perfect negative linear relationship.
- $r = 0$: No linear relationship.

Correlation and Linear Independence

Key Insight:

- If $r = 0$ (zero correlation), the variables are uncorrelated, but this does not imply independence unless the variables are normally distributed.
- Linear independence means that one variable cannot be expressed as a linear combination of another.

Geometric Interpretation:

- Variables X and Y can be treated as vectors in n -dimensional space.
- **Angle between vectors:** If θ is the angle between X and Y , the correlation coefficient is:

$$r = \cos(\theta)$$

- $\cos(\theta) = 0$: Orthogonal (uncorrelated).
- $\cos(\theta) = 1$: Perfectly aligned (positive correlation).
- $\cos(\theta) = -1$: Opposite direction (negative correlation).

Properties of Variance

- **Zero Variance:**

$$\text{Var}(X) = 0 \quad \text{if and only if} \quad X \text{ is constant.}$$

- **Scaling Property:**

$$\text{Var}(cX) = c^2 \cdot \text{Var}(X),$$

where c is a constant.

- **Shifting Property:**

$$\text{Var}(X + c) = \text{Var}(X),$$

where c is a constant.

- **Variance of the Sum/difference of Independent Random Variables:** If X and Y are independent random variables:

$$\text{Var}(X \pm Y) = \text{Var}(X) + \text{Var}(Y).$$

- **Variance of the Sum/difference of Dependent Random Variables:** If X and Y are dependent random variables:

$$\text{Var}(X \pm Y) = \text{Var}(X) + \text{Var}(Y) \pm 2\text{Cov}(X, Y).$$

Standard Deviation

Definition: The square root of variance, representing dispersion in the original units of data.

Formula:

$$\text{Standard Deviation} = \sigma = \sqrt{\text{Variance}} = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}}$$

Characteristics:

- Same units as the original data, making it easier to interpret.
- Measures average deviation from the mean.
- Sensitive to extreme values.

Example: Using the variance $\sigma^2 = 5$ (from the previous slide):

$$\text{Standard Deviation} = \sqrt{5} \approx 2.24$$

Relative Measures of Dispersion

Relative Range and Coefficient of Range

The **Range** and its variations are important measures of dispersion that help to understand the spread of data.

- **Relative Range:** The range normalized by the mean or the average of the data. It expresses how wide the range is relative to the mean.

$$\text{Relative Range} = \frac{\text{Range}}{\bar{X}} = \frac{\text{Max}(X) - \text{Min}(X)}{\bar{X}}$$

- **Range Coefficient:** A dimensionless measure of dispersion that is the ratio of the range to the sum of the maximum and minimum values.

$$\text{Range Coefficient} = \frac{\text{Max}(X) - \text{Min}(X)}{\text{Max}(X) + \text{Min}(X)}$$

These measures help in comparing the spread of data across different datasets.

Coefficient of IQR (Quartile Coefficient of Dispersion)

The **Coefficient of IQR** (also called the **Quartile Coefficient of Dispersion**) is a relative measure of dispersion that uses the **Interquartile Range (IQR)** to assess the spread of the middle 50% of the data.

- **Quartile Coefficient of Dispersion:** A dimensionless measure that expresses the IQR relative to the sum of the first and third quartiles.

$$\text{Quartile Coefficient of Dispersion} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

- **Interpretation:**

- A **high value** of the coefficient indicates a **larger spread** in the central 50% of the data.
- A **low value** suggests the data is more **concentrated** around the median.

The **Quartile Coefficient of Dispersion** is particularly useful for comparing the spread of different datasets, especially when the data contains outliers or is skewed.

Coefficient of Mean Deviation

Definition: The coefficient of mean deviation is a relative measure of dispersion that expresses the mean deviation as a proportion of a central value, allowing for comparisons across different datasets. It is calculated as:

$$\text{Coefficient of Mean Deviation} = \frac{\text{Mean Deviation}}{\text{Mean}} = \frac{1}{n\bar{x}} \sum_{i=1}^n |x_i - \bar{x}|$$

- It can be measured about the mean, median, or mode.
- It provides a clear percentage or proportion of deviation relative to the central value.

Coefficient of Variation (CV)

Definition: A standardized measure of dispersion expressed as a percentage of the mean.

Formula:

$$CV = \frac{\sigma}{\mu} \times 100\%$$

- Useful for comparing variability across datasets with different units or scales.
- Independent of the unit of measurement.

Example: Dataset: $\{10, 20, 30, 40, 50\}$ Mean: $\mu = 30$, Standard Deviation: $\sigma = 15.81$

$$CV = \frac{15.81}{30} \times 100\% \approx 52.7\%$$

Chebyshev's Rule: Overview

Chebyshev's Rule provides a universal bound on the proportion of data within a certain number of standard deviations (k) from the mean for any distribution.

Key Formula:

$$P(|X - \mu| < k\sigma) \geq 1 - \frac{1}{k^2}, \quad k > 1$$

Key Components:

- X : Random variable.
- μ : Mean of the distribution.
- σ : Standard deviation of the distribution.
- k : Number of standard deviations away from the mean.

Applicability: Works for any distribution, including those that are not normal.

Chebyshev's Rule: Interpretation

Interpretation for Specific k -values:

- For $k = 2$: At least 75% of data lies within 2σ .
- For $k = 3$: At least 88.89% of data lies within 3σ .
- For $k = 4$: At least 93.75% of data lies within 4σ .

Example: Consider a dataset with:

- Mean $(\mu) = 50$.
- Standard deviation $(\sigma) = 10$.

For $k = 2$:

$$P(|X - 50| < 2 \times 10) \geq 1 - \frac{1}{2^2} = 0.75$$

Conclusion: At least 75% of the data lies within $[30, 70]$.

Applications of Chebyshev's Rule

1. Non-Normal Distributions:

- Predicts data spread for distributions that are not bell-shaped.

2. Anomaly Detection:

- Identifies outliers as points lying far outside $k\sigma$.

3. Data Assurance:

- Validates the proportion of data within a reasonable range, even for unknown distributions.

Example:

- Data: Test scores with $\mu = 70$ and $\sigma = 15$.
- For $k = 3$:

$$P(|X - 70| < 3 \times 15) \geq 1 - \frac{1}{3^2} = 0.8889$$

- At least 88.89% of scores lie within $[25, 115]$.

Empirical Rule (68-95-99.7 Rule):

- The Empirical Rule applies to *normal distributions* and states the approximate percentages of data that lie within certain standard deviations from the mean.
- It is often referred to as the 68-95-99.7 rule because:
 - **68% of the data** lie within one standard deviation (σ) from the mean (μ).
 - **95% of the data** lie within two standard deviations from the mean.
 - **99.7% of the data** lie within three standard deviations from the mean.

Formula:

$$\mu - \sigma \leq x \leq \mu + \sigma \quad (68\% \text{ of the data})$$

$$\mu - 2\sigma \leq x \leq \mu + 2\sigma \quad (95\% \text{ of the data})$$

$$\mu - 3\sigma \leq x \leq \mu + 3\sigma \quad (99.7\% \text{ of the data})$$

Limitations:

- The Empirical Rule only applies to *bell-shaped, symmetric normal distributions*.
- For skewed or non-normal distributions, other methods (e.g., Chebyshev's inequality) should be used.

Empirical Rule vs. Chebyshev's Rule

Aspect	Empirical Rule	Chebyshev's Rule
Distribution Assumption	Applies only to normal (Gaussian) distributions	Applies to any distribution, including non-normal
Precision	Provides specific percentages (68%, 95%, 99.7%)	Provides general lower bounds (e.g., 75%, 89%)
Applicability	Limited to normal distributions	Broad applicability to any dataset
Data Spread	Gives exact data spread for normal distributions	Gives a minimum proportion for all datasets

Table: Comparison of Empirical Rule and Chebyshev's Rule

Definition of the Range Rule of Thumb

- The **Range Rule of Thumb** is a simple method to estimate the **standard deviation** (σ) of a dataset.

- Formula:

$$\sigma \approx \frac{\text{Range}}{4}$$

- Where:

- σ : Standard deviation
- **Range**: Difference between the maximum and minimum values of the dataset:

$$\text{Range} = \text{Max} - \text{Min}$$

Why $\frac{\text{Range}}{4}$?

- In a normal distribution:
 - Approximately 95% of the data lies within $\pm 2\sigma$ of the mean.
 - This means the range spans about 4σ :

$$\text{Range} \approx 4\sigma.$$

- Dividing the range by 4 provides a quick approximation of σ .

When the Rule Works Well and When the Rule May Fail

- The Rule Works Well
 - The dataset is approximately **normally distributed**.
 - There are no significant **outliers** to distort the range.
- The Rule May Fail
 - The data is **highly skewed** (e.g., J-shaped or L-shaped distributions).
 - The data contains extreme **outliers** that inflate the range.

Example

- Consider the dataset: 5, 8, 12, 15, 20.
- **Step 1: Calculate the Range**

$$\text{Range} = 20 - 5 = 15.$$

- **Step 2: Estimate σ**

$$\sigma \approx \frac{\text{Range}}{4} = \frac{15}{4} = 3.75.$$

Empirical Relation in Statistics

Empirical Relation:

- The *Empirical Relation* in statistics provides approximate relationships between the mean, median, and mode of a dataset, specifically for unimodal and symmetric distributions.
- It is useful for approximating the central tendency when the distribution of data is not fully known.

Common Empirical Relations:

$$\text{Mode} \approx 3 \times \text{Median} - 2 \times \text{Mean} \quad (1)$$

or equivalently,

$$\text{Mean} - \text{Mode} \approx 3(\text{Mean} - \text{Median}) \quad (2)$$

• Empirical Relation for a Symmetric Distribution:

$$\text{Mean} \approx \text{Median} \approx \text{Mode}$$

- **Empirical Relation for Skewed Distributions:**

Mean $>$ Median $>$ Mode (For positively skewed distributions)

Mean $<$ Median $<$ Mode (For negatively skewed distributions)

Explanation:

- For symmetric distributions like the normal distribution, the mean, median, and mode are approximately equal.
- For skewed distributions, the mean will be pulled in the direction of the skew.